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Kolmogorov-Smirnov Test For Two-Dimensional Data

William H. Press and Saul A. Teukolsky

How to tell whether a set of (x,y) data points are consistent with a particular probability distribution, or with another data set

The Kolmogorov-Smirnov (K-S) test is a standard statistical test for deciding whether a set of data is consistent with a given probability distribution. For example, the data set might be the list of values $x = 0.021$, 0.082, 0.143, 0.520, 1.788, and 4.712, and the probability distribution might be $p(x)dx = e^{-x}dx$. The classical K-S test is one-dimensional–each data point is a single number–and the above example is a one-sample test, where one data set is compared to an analytic model. There is also a similar two-sample K-S test, which has two data sets instead of a data set and a model. The two-sample test checks whether two data sets are both drawn from the same underlying probability distribution, but without assuming any specific model for that distribution. For example, the two-sample K-S test could be used to determine whether the list of values above could come from the same distribution as the list $x = 0.054$, 0.129, 0.384, 2.452. Note that our two data sets need not have the same size. In practice the K-S test would be done with a larger number of data points in each set, no less than of order 10 or 20.

These one-dimensional K-S tests are described in full detail in most statistics textbooks. While it would be redundant to give a complete discussion here, a quick summary of the mechanics of these one-dimensional K-S test will be a necessary preface to our main subject, which is a two-dimensional generalization of the K-S test.

Any model distribution function $p(x)$ has an associated cumulative distribution function

$$P(x) \equiv \int_{-\infty}^{x} p(s)ds.$$

Obviously, since probablity is normalized, this varies from 0 at $x = -\infty$ to 1 at $x = \infty$. A data set x_i, $i = 1...N$ also has a cumulative distribution function, namely the "staircase function" $S_N(x)$ defined as the fraction of the N data values to the left of x. Note that $S_N(x)$ starts at zero, jumps by the same constant $1/N$ at each x_i, and finishes at 1.

The Kolmogorov-Smirnov statistic is the maximum difference between the cumulative distribution function of the data and model (one-sample), or between the two data sets (two-sample), that is,

$$D = \max_{-\infty < x < \infty} |S_N(x) - P(x)| \tag{1}$$

$$D = \max_{-\infty < x < \infty} |S_{N_1}(x) - S_{N_2}(x)| \tag{2}$$

What makes the K-S statistic useful is that its distribution in the case of the null hypothesis (data sets

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drawn from the same distribution) can be calculated, giving the significance of any observed nonzero value of D. In particular, the calculation of the significance is universal, that is, independent of the shape of the distribution being scrutinized. (This follows essentially from the fact that any one-dimensional distribution can be reparametrized so as to make it uniform.)

A function which enters into the calculation of the significance can be written as the following sum,

$$Q_{KS}(\lambda) = 2\sum_{j=1}^{\infty}(-)^{j-1}e^{-2j^2\lambda^2} \tag{3}$$

which is a monotonic function with the limiting values

$$Q_{KS}(0) = 1 \qquad Q_{KS}(\infty) = 0 \tag{4}$$

In terms of this function, the significance level of an observed value of D (as a disproof of the null hypothesis that the distributions are the same) is given approximately by the formulas

$$\text{Probability}\,(D > \text{observed}) = Q_{KS}(\sqrt{N}D) \tag{5}$$

for the case of one sample, where N is the number of data points, and

$$\text{Probability}\,(D > \text{observed}) = Q_{KS}\left(\sqrt{\frac{N_1N_2}{N_1+N_2}}D\right) \tag{6}$$

for the case of two samples, where N_1 is the number of data points in the first, N_2 the number in the second. A small value of the above probability indicates that the data set and the model, or the two data sets, are significantly different.

The *Numerical Recipes* books give computer implementations of the one- and two-sample K-S tests. The routine KSONE takes as input an array DATA of length N and a user-supplied function FUNC (the cumulative distribution function $P(x)$ for the model), and returns the K-S statistic D and its probability (significance) PROB. Similarly, the routine KSTWO takes as input arrays DATA1 of length N1 and DATA2 of length N2, and returns D and PROB. These routines use a function PROBKS which directly calculates equation (3), and which we will make use of below.

So much for one dimension. We now move into less familiar territory, and discuss a highly useful generalization of the K-S test to two-dimensional distributions. This generalization is due to G. Fasano and A. Franceschini [*Mon. Not. Roy. Astr. Soc.*, 225, 155 (1987)] and extends an earlier idea due to J. A. Peacock, [*Mon. Not. Roy. Astr. Soc.*, 202, 615 (1983)].

In a two-dimensional distribution, each data point is characterized by an (x,y) pair of values. An example near to our hearts is that each of the 19 neutrinos which were detected from Supernova 1987A is characterized by a time t_i and by an energy E_i (see D. N. Spergel et al. [*Science*, 237, 1471 (1987)]). We might wish to know whether these measured pairs (t_i,E_i), $i = 1...19$ are consistent with a theoretical model that predicts neutrino flux as a function of both time and energy–that is, a two-dimensional probability distribution in the (x,y) [here, (t,E)] plane. That would be a one-sample test. Or, given two sets of neutrino detections, from two comparable detectors, we might want to know whether they are compatible with each other, a two-sample test.

In the spirit of the tried and true one-dimensional K-S test, we want to range over the (x,y) plane in search of some kind of maximum cumulative difference between two two-dimensional distributions. Unfortunately, cumulative probability distribution is not well-defined in more than one dimension! Peacock's insight was that a good surrogate is the integrated probability in each of four natural

```
SUBROUTINE KS2D1S(X1,Y1,N1,QUADVL,D1,PROB)
  Two-dimensional Kolmogorov-Smirnov test of one sample against a model. Given the x
  and y coordinates of N1 data points in arrays X1 and Y1, and given a user-supplied function
  QUADVL that exemplifies the model, this routine returns the 2-dimensional K-S statistic as
  D1, and its significance level as PROB. Small values of PROB show that the sample is
  significantly different from the model. Note that the test is slightly distribution-dependent,
  so PROB is only an estimate.
DIMENSION X1(N1),Y1(N1)
D1=0.0
DO11 J=1,N1                          Loop over the data points.
  CALL QUADCT(X1(J),Y1(J),X1,Y1,N1,FA,FB,FC,FD)
  CALL QUADVL(X1(J),Y1(J),GA,GB,GC,GD)
  D1=MAX(D1,ABS(FA-GA),ABS(FB-GB),ABS(FC-GC),ABS(FD-GD))
  For both the sample and the model, the distribution is integrated in each of four quadrants, and the
  maximum difference is saved.
11CONTINUE
CALL PEARSN(X1,Y1,N1,R1,DUM,DUMM)          Get the linear correlation coefficient R1.
SQEN=SQRT(FLOAT(N1))
RR=SQRT(1.0-R1**2)
Estimate the probability using the K-S probability function PROBKS.
PROB=PROBKS(D1*SQEN/(1.0+RR*(0.25-0.75/SQEN)))
RETURN
END
```

```
SUBROUTINE QUADVL(X,Y,FA,FB,FC,FD)
  This is a sample of a user-supplied routine to be used with KS2D1S. In this case, the
  model distribution is uniform inside the square -1 < x < 1, -1 < y < 1. In general this
  routine should return, for any point (X,Y), the fraction of the total distribution in each of
  the four quadrants around that point. The fractions, FA, FB, FC, and FD, must add up to 1.
  Quadrants are alphabetical, counterclockwise from the upper-right.
QA=MIN(2.,MAX(0.,1.-X))
QB=MIN(2.,MAX(0.,1.-Y))
QC=MIN(2.,MAX(0.,X+1.))
QD=MIN(2.,MAX(0.,Y+1.))
FA=0.25*QA*QB
FB=0.25*QB*QC
FC=0.25*QC*QD
FD=0.25*QD*QA
RETURN
END
```

quadrants around a given point (x_i, y_i), namely the total probabilities (or fraction of data) in $(x > x_i, y > y_i)$, $(x < x_i, y > y_i)$, $(x < x_i, y < y_i)$, $(x > x_i, y < y_i)$. The two-dimensional K-S statistic D is now taken to be the maximum difference (ranging both over data points and over quadrants) of the corresponding integrated probabilities. (In the case of two samples, the statistic D is the mean of the two values obtained when (x_i, y_i) ranges over the two data sets.) If you are confused at this point about the exact definition of D, don't fret; the accompanying computer routines amount to a precise algorithmic definition.

Figure 1 gives a feeling for what is going on. The 65 triangles and 35 squares seem to have somewhat different distributions in the plane. The dotted lines are centered on the triangle that maximizes the D statistic; the maximum occurs in the upper-left quadrant. That quadrant contains only 0.12 of all the triangles, but it contains 0.56 of all the squares. The value of D is thus 0.44. Is this statistically significant? Read on.

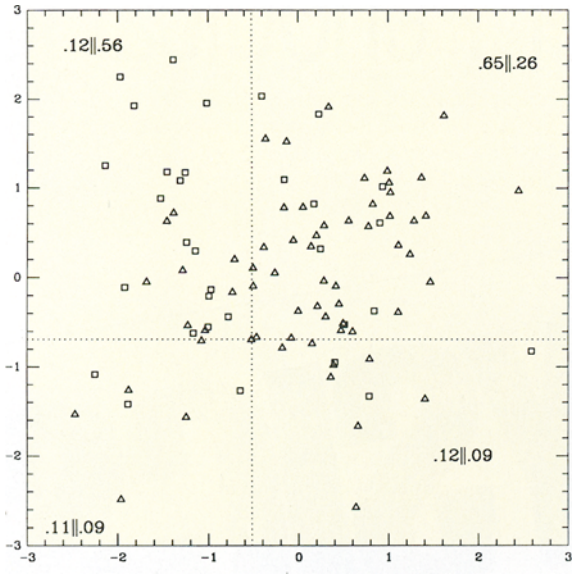
Even for fixed sample sizes, it is unfortunately not rigorously true that the distribution of D in the null hypothesis is independent of the shape of the two-dimensional distribution. In this respect the two-dimensional K-S test is not as "neat" as its one-dimensional parent. However, extensive Monte Carlo integrations have shown that the distribution of the two-dimensional D is very nearly identical for even quite different distributions, as long as they have the same coefficient of correlation r, defined in the usual way by

$$r = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2} \sqrt{\sum_i (y_i - \bar{y})^2}} \tag{7}$$

In their paper, Fasano and Franceschini tabulate Monte Carlo results for (what amounts to) the distribution of D as a function of–of course–D, sample size N, and coefficient of correlation r. Analyzing their results, we have found that the significance levels for the two-dimensional K-S test can be summarized by the simple, though approximate, formulas,

```
SUBROUTINE KS2D2S(X1,Y1,N1,X2,Y2,N2,D,PROB)
  Two-dimensional Kolmogorov-Smirnov test on two samples. Given the x and y coordinates
  of the first sample as N1 values in arrays X1 and Y1, and likewise for the second sample, N2
  values in arrays X2 and Y2, this routine returns the 2-dimensional, 2-sample K-S statistic
  as D, and its significance level as PROB. Small values of PROB show that the two samples
  are significantly different. Note that the test is slightly distribution-dependent, so PROB is
  only an estimate.
DIMENSION X1(N1),Y1(N1),X2(N2),Y2(N2)
D1=0.0
DO[11] J=1,N1                          First, use points in the first sample as origins.
   CALL QUADCT(X1(J),Y1(J),X1,Y1,N1,FA,FB,FC,FD)
   CALL QUADCT(X1(J),Y1(J),X2,Y2,N2,GA,GB,GC,GD)
   D1=MAX(D1,ABS(FA-GA),ABS(FB-GB),ABS(FC-GC),ABS(FD-GD))
  [11]CONTINUE
D2=0.0
DO[12] J=1,N2                          Then, use points in the second sample as origins.
   CALL QUADCT(X2(J),Y2(J),X1,Y1,N1,FA,FB,FC,FD)
   CALL QUADCT(X2(J),Y2(J),X2,Y2,N2,GA,GB,GC,GD)
   D2=MAX(D2,ABS(FA-GA),ABS(FB-GB),ABS(FC-GC),ABS(FD-GD))
  [12]CONTINUE
D=0.5*(D1+D2)                          Average the K-S statistics.
SQEN=SQRT(FLOAT(N1)*FLOAT(N2)/FLOAT(N1+N2))
CALL PEARSN(X1,Y1,N1,R1,DUM,DUMM)      Get the linear correlation coefficient for
CALL PEARSN(X2,Y2,N2,R2,DUM,DUMM)      each sample.
RR=SQRT(1.0-0.5*(R1**2+R2**2))
Estimate the probability using the K-S probability function PROBKS.
PROB=PROBKS(D*SQEN/(1.0+RR*(0.25-0.75/SQEN)))
RETURN
END
```

```
SUBROUTINE QUADCT(X,Y,XX,YY,NN,FA,FB,FC,FD)
  Given an origin (X,Y), and an array of NN points with coordinates XX and YY, count how
  many of them are in each quadrant around the origin, and return the normalized frac-
  tions. Quadrants are labeled alphabetically, counterclockwise from the upper-right. Used
  by KS2D1S and KS2D2S.
DIMENSION XX(NN),YY(NN)
NA=0
NB=0
NC=0
ND=0
DO[12] K=1,NN
  IF(YY(K).GT.Y)THEN
    IF(XX(K).GT.X)THEN
      NA=NA+1
    ELSE
      NB=NB+1
    ENDIF
  ELSE
    IF(XX(K).GT.X)THEN
      ND=ND+1
    ELSE
      NC=NC+1
    ENDIF
  ENDIF
 [12]CONTINUE
FF=1.0/NN
FA=FF*NA
FB=FF*NB
FC=FF*NC
FD=FF*ND
RETURN
END
```



1. Two-dimensional distributions of 65 triangles and 35 squares. The two-dimensional Kolmogorov-Smirnov test finds that point one of whose quadrants (shown by dotted lines) maximizes the difference between fraction of triangles and fraction of squares. Then, equation (8) indicates whether the difference is statistically significant, i.e., whether the triangles and squares must have different underlying distributions.

Probability ($D >$ observed) =

$$Q_{KS}\left(\frac{\sqrt{N}D}{1+\sqrt{1-r^2}(0.25-0.75/\sqrt{N})}\right) \tag{8}$$

for the one-sample case, and the same for the two-sample case, but with

$$N=\sqrt{\frac{N_1N_2}{N_1+N_2}}. \tag{9}$$

The above formulas are accurate enough when $N \gtrsim 20$, and when the indicated probability (significance level) is less than (more significant than) 0.20 or so. When the indicated probability is >0.20, its value may not be accurate, but the implication that the data and model (or two data sets) are not significantly different is certainly correct. In fact, few investigators would claim significance for a result unless the probability is less than 0.05 or 0.01. In this regime, equations (8) and (9) are always pretty good. Notice that in the limit of $r \to 1$ (perfect correlation), equations (8) and (9) reduce to equations (5) and (6): the two-dimensional data lie on a perfect straight line, and the two-dimensional K-S test goes neatly over to a one-dimensional K-S test.

The significance level for the data in Figure 1, by the way, is about 0.001. This establishes to a near-certainty that the triangles and squares were drawn from different distributions, as in fact they were.

Of course, if you are uncomfortable with the "horseback approximation" of equation (8), you have the alternative of doing your own Monte Carlo. Use equation (8) only as a rough guide to find the cases of interest. Then, generate a lot of synthetic data sets from your model, each one with the same number of points as the real data set. Compute D for each synthetic data set, using the accompanying computer routines (but ignoring their calculated probabilities), and count what fraction of the time these synthetic D's exceed the D from the real data. That fraction is your significance. If it differs by more than a factor 1.5 from equation (8), we would like to know about it!

We don't know any way to do Monte Carlo without a model, so we don't know any way to check on equation (9) if you are doing a two-sample test. You have to take it, or leave it.

One disadvantage of the two-dimensional tests, by comparison with their one-dimensional progenitors, is that the two-dimensional tests require of order N^2 operations: Two nested loops of order N take the place of an $N \log N$ sort. For small computers, this restricts the usefulness of the tests to N less than a few thousand.

It remains only to discuss the accompanying computer implementations. The one-sample case is embodied in the routine KS2D1S (that is, 2-dimensions, 1-sample). This routine calls a straightforward utility routine QUADCT to count points in the four quadrants, and it calls a user-supplied routine QUADVL which must be capable of returning the integrated probability of an analytic model in each of four quadrants around an arbitrary (x,y) point. A trivial sample QUADVL is shown; realistic QUADVLs can be quite complicated, often incorporating numerical quadratures over analytic two-dimensional distributions.

The routine KS2D1S calls on two short routines from the *Numerical Recipes* books, PEARSN, used only to evaluate equation (7), and PROBKS, which evaluates equation (3). If you don't have *Numerical Recipes*, equivalent routines can be found in most mathematical subroutine libraries, or you can very easily write your own from the equations given above.

The routine KS2D2S is the two-sample case of the two-dimensional K-S test. It also calls QUADCT, PEARSN, and PROBKS. Being a two-sample test, it does not need an analytic model.

FOR FURTHER READING

Numerical Recipes: The Art of Scientific Computing, by William H. Press, Brian P. Flannery, Saul A. Teukolsky, and William T. Vetterling (Cambridge Univerisy Press, 1986) and the just-released *Numerical Recipes in C*, by the same authors, provides the background for this series of columns. Cambridge University Press is offering a 20% discount on the books to *CIP* readers. To order, call (800) 872-7423 (outside NY state), or (800) 227-0247 (within NY state), and please mention *CIP* when ordering. ■

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